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Paleomagnetism of Late Cretaceous Alkaline Rocks in the Nemuro Peninsula, Hokkaido, N.E. Japan

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A paleomagnetic study was made on 747 specimens from 57 sites from Late Cretaceous alkaline rocks in the Nemuro Peninsula, Hokkaido, N.E. Japan (43.3°N, 145.8°E). Direction of magnetization of samples from the marginal parts of some monzonite or piclitic dolerite complex are quite different from those from the central parts. This suggests that the transitional change of geomagnetic field polarity might have occurred during cooling of the rock body. The paleomagnetic results suggest the presence of two short reversed zones in the Upper Cretaceous in which normal polarity was predominant. Recent K-A age determinations suggest that these reversals took place at 88 m.y. and 84 m.y. ago. They should become important time markers in Cretaceous paleomagnetic stratigraphy.

1. Introduction

Alkaline rock complexes such as sheets and sills as well as lava flows occur widely in various stratigraphic horizons in the Upper Cretaceous deposits in the Nemuro Peninsula, situated in the easternmost part of Hokkaido, N.E. Japan. Due to the remarkable differentiated rock facies, from porphyritic basalt to monzonite, and their peculiar chemical composition, they were the subject of many petrographic studies (SUZUKI, 1938; YAGI, 1958, 1968, etc.). We reported the paleomagnetic results of these rocks (FUJIWARA and NAGASE, 1965). Since then, further field and laboratory works and recent K-A age determinations have brought up some new aspects in terms of the age and activity of alkaline rocks in the Nemuro Peninsula. The results of paleomagnetic determinations on newly collected samples and their significance as to Late Cretaceous geomagnetic polarity changes are briefly given in this paper.

2. Measurements

Generally about 6 oriented samples were obtained from each sampling site.

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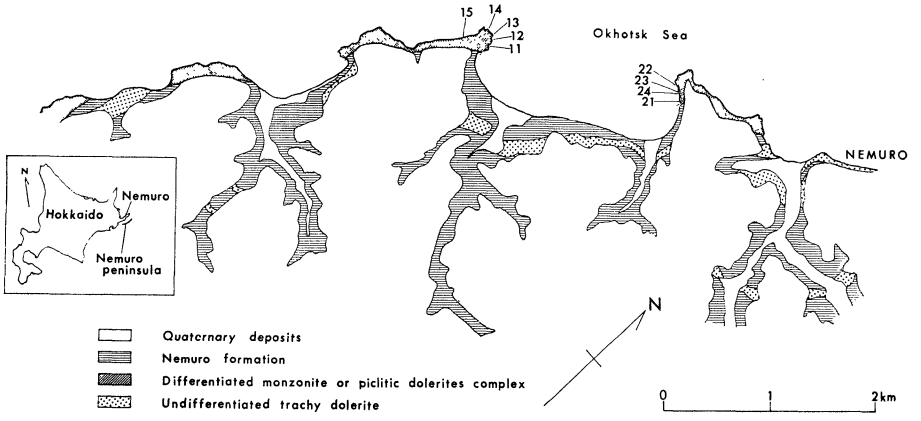


Fig. 1. Brief geological sketch map showing the sampling localities of Poromoshiri complex (left) and Honioi complex (right).

Several cylindrical specimens (3 cm in diameter) were drilled from one sample resulting in 747 specimens from 57 sites. The sampling localities are shown in Fig. 1.

Measurements of the intensity and direction of magnetization have been made using an astatic magnetometer. The mean direction of remanent magnetization and statistical parameters of each site have been obtained using the method of FISHER (1953). Both alternating field demagnetization and thermal demagnetization have been used to test the stability of NRM of each group of samples. Thermomagnetic analysis for at least one specimen from each sampling site, has been carried out to investigate the change of saturation magnetization with temperature and the Curie temperature of the magnetic minerals using an automatic thermomagnetic balance.

Intensities of NRM collected from trachy dolerite with pillow structure and the marginal part of the piclitic dolerite and monzonite complex are extremely high. As shown in Table 1, intensities of NRM of the pillow part range between 1.0 and 1.3×10^{-2} emu/gr. The NRM intensities of the marginal part of the piclitic dolerite and monzonite complex range between 1.0 and 7.0×10^{-3} emu/gr., while those for the monzonitic part of differentiated dolerite complex fall between 2.0 and 9.8×10^{-4} emu/gr. These discrepancies might be explained in terms of their different grain sizes of ferromagnetic minerals under the different cooling conditions.

The results of thermal demagnetization of one or two specimens from each site do not show any remarkable scattering in the direction of magnetization during heat treatment in the temperature range 100°–500°C.

The results of paleomagnetic measurements of 411 specimens from 44 sites are tabulated in Table 1. Trachy dolerites with pillow structures are reversely magnetized while undifferentiated trachy dolerites have normal directions. The

Table 1.

Site no.	N	N.R.M.			After cleaning					Curie temperature (°C)	Pole position
		$D°$	$I°$	$\alpha_{95}°$	M	$D°$	$I°$	$\alpha_{95}°$	$\tilde{H}$(oe)		
Differentiated monzonite or piclitic dolerite complex											
Poromoshiri Complex											
11	23	308	45	10	52	305	47	7	200	567	
12	18	265	55	15	7.5	318	45	7	200	570	
13	2	342	60	9	0.4	340	59	5	100		
14	33	325	52	6	7.4	328	57	4	100	570	
15	2	242	60	30	0.2	281	50	22	300		
Mean (11, 12, 14)						325	54	2			62N, 051E
Honioi Complex											
21	23	105	−70	15	5.7	107	−65	11	300	550	
22	33	096	−62	15	5.3	114	−60	5	100	545	
23	14	113	−44	18	4.9	120	−60	7	200	565	
Mean (21–23)						114	−62	3			45S, 096W
24	9	330	20	18	7.4	317	03	3	300		33N, 020E
Nosappu Complex											
26	7	010	65	26	1.6	355	72	10	200	570	
27	7	350	72	5	4.5	006	68	3	100	570	
28	8	323	68	18	4.8	344	56	6	100	570	
29	8	263	45	15	6.9	339	55	5	100	555	
30	6	003	56	5	3.8	003	56	5	50	565	
31	12	010	72	18	3.4	353	61	10	300	545	
32	5	340	15	10	2.2	344	26	12	100	560	
33	6	335	07	12	2.9	352	11	8	200	545	
34	8	349	54	10	3.4	346	61	7	200	555	
35	5	003	60	15	2.9	349	54	10	200	555	
36	8	320	36	22	4.0	349	48	4	300	545	
37	12	358	52	18	3.2	351	46	10	300	550	
38	8	356	22	20	3.5	355	46	10	100	550	
Mean (26–38)						350	51	5			76N, 003E
39	8	204	68	32	2.6	194	44	11	300	530	29S, 118E
40	7	205	30	12	2.4	205	37	9	200	520	20S, 133E
41	12	058	−35	15	3.4	048	−43	7	200	540	09N, 076W
42	9	008	−25	20	2.6	330	−47	7	200	560	14E, 007W
43	6	303	40	15	2.1	291	43	11	200	530	32N, 060E
44	6	315	52	12	1.9	314	44	10	100	540	49N, 014W
45	4	287	42	20	3.9	351	35	8	200	550	65N, 014W
46	6	340	48	6	7.7	336	46	5	100	550	
47	6	015	62	18	8.4	351	49	3	200	570	
48	6	010	57	2	5.0	010	57	2	100		
49	6	355	63	16	5.0	016	51	3	200		
50	4	013	50	3	5.7	018	54	2	50		
51	7	020	60	4	5.5	020	62	2	50		

(continued)

Table 1. (continued)

Site no.	N	N.R.M.			After cleaning					Curie temperature (°C)	Pole position
		$D°$	$I°$	$\alpha_{95}°$	M	$D°$	$I°$	$\alpha_{59}°$	$\tilde{H}$(oe)		
52	7	359	51	2	5.8	359	51	2	50		
53	6	000	47	2	10	000	48	2	50		
54	6	352	50	4	5.1	001	52	4	100		
55	6	001	55	3	9.4	348	55	2	50		
Mean (46-55)						000	53	2			81N, 034W
		Undifferentiated trachy dolerite									
2	10	314	83	9	2.6						
7	14	004	77	17	2.0						
Mean		347	81	7							54N, 124E
		Trachy dolerite with pillow structure									
3	6	136	−66	15	110						
10	6	171	−49	20	116						
Mean		158	−59	12							74N, 092W

Note: N, number of specimen measured; D, declination; I, inclination; M, intensity of magnetization ($\times 10^{-4}$ emu/gr); α_{95}, radius of the circle of 95% confidence in degree of arc; $\tilde{H}$, peak field intensity in AC washing.

magnetization of differentiated monzonite or piclitic dolerite complexes is rather complicated. The samples were collected from three different complexes. Of these, the Nosappu complex* contains two different magnetic components. The marginal facies which comprises porphyritic basalt or piclitic dolerite has a direction of 350°, +53° in declination and inclination with $\alpha_{95}=5°$, inferred from 100 specimens from sites 26–38, while samples collected from the monzonitic part in the central part of sill have quite different directions as shown in Table 1 and Fig. 2.

On the other hand, the marginal part of the Honioi complex* is reversely magnetized with the mean direction of 114°, −62° in declination and inclination, and $\alpha_{95}=3°$ based on 70 specimens from site 21, 22, 23 while 9 specimens of site 24 which were collected from the central part of this complex show normal polarity. The Poromoshiri complex* is magnetized normally. The mean direction is calculated as 325°, +54° in declination and inclination, and $\alpha_{95}=2°$ on basis of 78 specimens from five sites.

3. Discussion

The intrusions of the alkaline rocks in the Nemuro Peninsula can be divided into two stages from field observations (MITANI *et al.*, 1958; HASEGAWA and MITANI, 1959; FUJIWARA and MITANI, 1959). The result of K-A age determinations supports these subdivisions (UEDA and AOKI, 1968). The undifferenti-

* see appendix.

ated trachy dolerites and trachy dolerites with pillow structure might belong to the first stage and the differentiated monzonite or piclitic dolerite complexes to the later stage. The present paleomagnetic results also suggest the occurrence of two successive igneous activities on the basis of the presence of two different paleomagnetic polarities. However, both normal and reverse polarities were determined from one complex of monzonite or piclitic dolerites. Three complexes of monzonite or piclitic dolerites had probably intruded continuously at the transition time of the geomagnetic field reversal. Namely, the existence of two different paleomagnetic polarities in one rock complex may be explained in terms of the assumption that the geomagnetic field had been normal at the time when the rock at the marginal part was cooled through the Curie temperature of the ferromagnetic minerals in the rock, and the field had reversed in polarity when the rock inside the intrusive body became magnetized during cooling.

The directions of that portion of the Nosappu complex which possesses normal polarity may correspond to the directions of the Poromoshiri complex which also possesses normal polarity. The reversely magnetized inner part of the Nosappu complex may correspond to the marginal part of the Honioi complex which has reversely magnetized components as shown in schematically in Fig. 2.

Therefore, the proposed igneous activity with time may be the following: Poromoshiri complex, Nosappu complex and Honioi complex in ascending order.

The assumption is also supported by the investigations of relative pole positions from each complex. Pole positions determined from the Nosappu complex are distributed in both hemispheres. Furthermore, their distributions seem to gradually move from the northern to the southern hemisphere along some specific path. In addition, pole positions from two other complexes, Poromoshiri and Honioi complexes, are situated approximately along the polar moving path obtained from the Nosappu complex (Fig. 3). The geomagnetic field in the Late Cretaceous, therefore may be considered to have been reversed with its axis

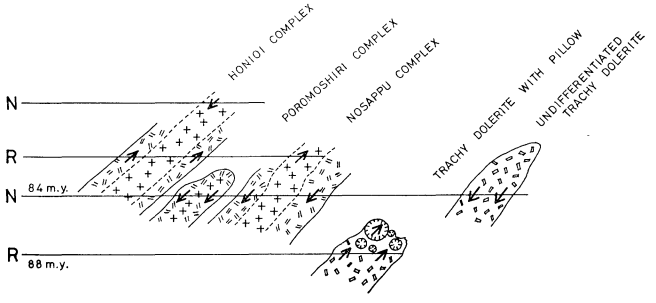


Fig. 2. Schematic representation of the sequence of the alkaline rocks in the Nemuro Peninsula. Horizontal line indicate the contemporaneity of the age of acquisition of remanent magnetization. N, normal polarity; R, reversed polarity.

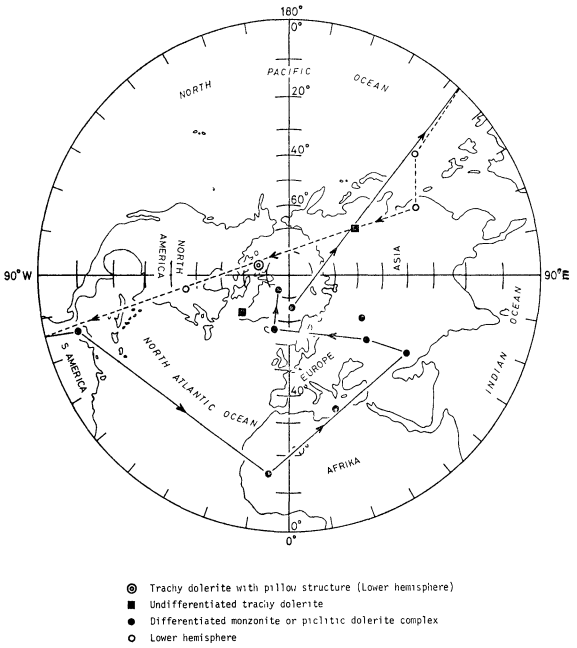


Fig. 3. Pole positions obtained from the alkaline rocks in the Nemuro Peninsula and the trace of migrating pole positions deduced from the Nosappu complex.

moving along a certain route as already many authors suggested (SIGURGEIRSSON, 1957; ITO, 1965, 1970; WATKINS, 1965; MOMOSE, 1958, 1963; BUCHA, 1970). Some continuous trace of the transitional field during a geomagnetic field reversal were reported. For instance, ITO (1970) suggested that the route of the pole during transition of the geomagnetic field in the Mio-Pliocence time may have followed a great circle on both hemispheres from about 10°E to 170°W in longitude. A detailed trace of a gradual reversal of the geomagnetic field from a series of Miocene lava flows in Oregon, U.S.A. was reported by WATKINS (1965). SIGURGEIRSSON (1957) reported four different paths obtained from four groups of lava flows in Iceland. Some part of their polar moving path during field transition seems to resemble the path obtained from the present study although the geologic ages in both cases are quite different.

The transitions of the geomagnetic field in the past seem to have occurred in a very short interval of time as compared with the length of the geological time. The time required for the transition in Miocene to Pliocene time is estimated to be of the order of 10^4 to 10^5 years and that in Pliocene to Pleistocene was of the order of 10^3 to 10^4 years on the basis of stratigraphic correlations of successive lava flows (ITO, 1970), depositional rates of clay sediments in U.S.S.R. (PETROVA, 1965) and combined results from worldwide paleomagnetic data (COX and DOELL, 1960). The present results also suggest that the transitional

field reversals also occurred in the Late Cretaceous and the time required for the field transition was probably short on the basis of the assumption that the field reversal probably occurred during the cooling process of the relatively thin intrusive body.

Many authors suggest a long interval of time of predominantly normal geomagnetic field polarity in the Late Cretaceous (SASAJIMA and SHIMADA, 1966; HELSLEY and STEINER, 1968; MCELHINNY and BUREK, 1971; LARSON and PITMAN, 1972; PECHERSKY and KHRAMOV, 1973; IRVING and COUILLARD, 1973). The geologic time ranges of the normal interval suggested by these authors differ to some extent as represented in Fig. 4, 77–102 m.y. (Helsley and Steiner), 70–120 m.y. (McElhinny and Burek), 84.5–111.5 m.y. (Larson and Pitman), 75–105 m.y. (Pechersky and Khramov) and 82–109 m.y. (Irving and Couillard) respectively. This normal interval was named the Mercanton Interval (Helsley and Steiner) or Jalal Interval (Pechersky and Khramov). According to MCELHINNY and BUREK (1971), this normal interval includes two short reversed zones which may become important global time markers. The presence of reversed zones in the Late Cretaceous was also proposed by SASAJIMA and SHIMADA (1966) and PECHERSKY and KHRAMOV (1973). The reversed zone was placed in the Middle Santonian (80 m.y. ±) on the basis of the stratigraphic correlation was called Akoh zone (SASAJIMA and SHIMADA, 1966; HELSLEY and STEINER, 1968; MCELHINNY and BUREK, 1971). PECHERSKY and KHRAMOV (1973) suggested the presence of a reversed zone (Kuldjin zone) in the Jalal normal Interval. The present results also suggest the presence of two reversals

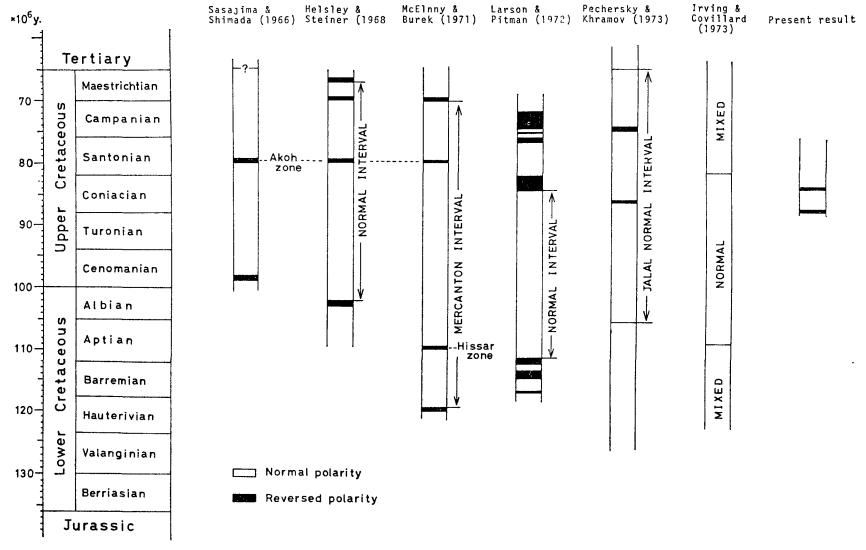


Fig. 4. Correlation diagram of reversal time scale in the Cretaceous. The geologic time scale is based on Holmes' symposium (1963).

of age about 88 m.y. and 84 m.y. The ranges of the normal interval and the short reversed zones in this interval have compiled in Fig. 4.

Discrepancies of age estimates between these results may probably arise from different methods of age determination and the difficulties in the stratigraphic correlations of local stratigraphic units to the international standard.

4. Conclusions

1) Alkaline rocks in the Nemuro Peninsula should be divided into two stages on the basis of paleomagnetic results.

2) The intrusions of monzonite or piclitic dolerite probably occurred at three different times. The results of paleomagnetic investigations of each complex suggest their intrusions might be continuous.

3) Presence of two different polarities in one rock complex may suggest that the geomagnetic field reversed during cooling. Analysis of distributions of the pole positions inferred from the samples collected from the Nosappu complex may suggest that the transitional geomagnetic field might have occurred in the Late Cretaceous.

4) The present results may suggest at least two short reversed periods in the long normal interval in the middle of Upper Cretaceous. These short reversed periods may become important time markers for Cretaceous paleomagnetic stratigraphy.

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APPENDIX: Stratigraphy and Petrographic Nature of Sampled Rocks

The Late Cretaceous deposits in the Nemuro Peninsula, about 4,500 m in total thickness, were called the Nemuro Group (SASA, 1934). It can be divided into five formations on the basis of their lithologic nature. The Notsukamappu, Nemuro, Choboshi, Ochiishi and Ururi Formation in ascending order (MITANI *et al.*, 1958; HASEGAWA and MITANI, 1959; FUJIWARA and MITANI, 1959).

ASANO (1962) reported the age of the Choboshi Formation probably to be Danian from the presenceof foraminiferas *Globorotalia triloclinoides* and *G. pseudobulloides*.

The alkaline rock in question, can be divided into 3 groups from their petrographic nature and occurrence: (1) differentiated monzonite or piclitic

dolerite complex, (2) undifferentiated trachy dolerite, and (3) trachy dolerite with pillow structure.

Monzonite or piclitic dolerite complexes of which the marginal part show sometimes consist of porphyritic basalt were typically developed near Nemuro city and Cape Nosappu. These rocks intrude into both Notsukamappu and Nemuro Formation as sills. The age of intrusion had been considered to be pre-Choboshi stage from the presence of conglomerate bearing piclitic dolerite pebbles in the Choboshi Formation. According to Ueda and Aoki (1968), the age of the monzonite or piclitic dolerite complexes can be considered to be 84 m.y. while that of the trachy dolerite with pillow structure is 88 m.y. on the basis of K-A age determinations on 6 specimens in the Nemuro Peninsula.

Undifferentiated trachy dolerites which intrude into both the Notsukamappu and Nemuro Formation are typically developed along the north coast of this peninsula. Thickness of these sheets and sills are generally observed to be less than 100 m. Marginal parts of these dolerites sometimes show porphyritic basalt texture.

Trachy dolerite with pillow structure is most recognizable at the Cape Hanasaki, westernmost part of this peninsula. This type of trachy dolerite can be observed in more than 10 stratigraphical horizons, from the basal part of the Notsukamappu Formation up to the upper part of the Nemuro Formation. In this paper, the complex of sites 21–24 is called the Honioi Complex for convenience. The other two are called the Poromoshiri complex (sites 11–15) and the Nosappu complex (sites 26–55).

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